
Evidence Before Ideation: A Literature-Review Infrastructure for Grounded Cross-Domain Research Ideas

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Abstract

Scientific assistants increasingly retrieve papers and synthesize related work, but the later ideation step is often delegated to free-form prompting that drops provenance. We argue that literature-review infrastructure should continue one step further: from evidence collection to grounded ideation. PaperNexus instantiates this view as a typed graph over paper-centered review state and a challenge-aware `idea_catalyst` workflow that returns staged packet bundles containing bridge paths, source takeaways, evidence spans, and evidence-backed idea fragments rather than only final suggestions. We update the evaluation around two deliberately separate questions. On public literature-retrieval benchmarks, PaperNexus’ native lightweight fixed-corpus scorer is plausible but not state of the art: on LitSearch it reaches 0.3739 nDCG@10 and 0.7165 Recall@100, below a BGE-small dense retriever; on full ScholarGym it reaches 0.1468 nDCG@10 and 0.3559 Recall@100, below a Qwen3-Embedding-0.6B dense artifact. The same evaluator can nevertheless ingest dense, fusion, and reranking artifacts: hybrid reranking raises LitSearch MRR@10 to 0.4211, while ScholarGym shows a useful cautionary result in which dense retrieval is strongest and fusion/reranking do not automatically help. On a six-case ideation audit, source-span packet availability is now satisfied by both a refreshed retrieve-then-prompt baseline and PaperNexus, so the remaining claim depends on strict labels for groundedness, unsupported bridge claims, and usable yield. The contribution is therefore an evidence-preserving ideation workflow and an honest audit protocol, not a retrieval-SOTA claim or a completed human evaluation.

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1. Introduction

Recent systems for scientific assistance are strongest at retrieval, synthesis, and question answering over papers (Lála et al., 2023; Asai et al., 2024). Benchmarks have accordingly emphasized literature search, discovery, and evidence lookup quality (Ajith et al., 2024; Shen et al., 2026; Xiong et al., 2026). Yet an important cognitive step in research still sits awkwardly between review and generation: given a target-domain challenge, how should an AI system propose cross-domain ideas without severing its links to evidence?

This gap matters for ML and CS workflows because early-stage ideation is rarely a pure brainstorming task. Researchers move from a problem statement, to unresolved challenges in the target domain, to potentially transferable mechanisms from other domains, and then to candidate hypotheses that must still be checked against the literature. Systems that stop at summary force this step back onto the user. Systems that jump directly to solution generation risk unsupported bridge claims. Recent work on interdisciplinary inspiration has made this ideation step explicit (Kargupta et al., 2026); parallel work on structured scientific memory shows that graph-backed research state can support more inspectable reasoning (Wu et al., 2026). We combine these directions.

This places the task in a specific middle ground. It is narrower than autonomous science, because the system is not expected to discover, validate, and execute research end to end. It is also stronger than basic literature search, because the target output is not a list of papers or a related-work summary, but a candidate idea whose supporting bridge claims remain auditable.

Our claim is narrow: literature-review infrastructure should support *grounded ideation*. In PaperNexus, ideation is not a free-form downstream prompt. It is a continuation of literature review over committed graph state. The system builds typed review artifacts such as `ResearchQuestion`, `Challenge`, `AbstractMechanism`, `Takeaway`, `IdeaFragment`, and `EvidenceSnippet`; the graph-mode `idea_catalyst` workflow then uses these structures to produce either evidence-backed idea fragments or an explicit requisition when the graph is too weakly

supported.

The task contract is correspondingly explicit. The input is a target-domain challenge interpreted over committed review state; the output is either (i) evidence-backed idea fragments with auditable bridge claims or (ii) a requisition that declares the missing evidence needed before a usable fragment can be surfaced. This is narrower than open-ended brainstorming and stronger than retrieval-only assistance.

This workshop paper makes three contributions. First, it frames grounded ideation as a literature-review task rather than a standalone creativity prompt. Second, it describes a graph-native workflow that preserves intermediate evidence objects, not only final ideas. Third, it reports the current benchmark boundary: PaperNexus can be compared to public retrieval systems at the retrieval layer, but its distinctive claim must be evaluated separately as evidence-contract preservation and idea audibility. The updated results are intentionally conservative: native lexical retrieval is weaker than dense retrieval on both LitSearch and ScholarGym; hybrid fusion and reranking help on LitSearch but not on ScholarGym; and the ideation table still requires independent labels before it can support a final idea-quality claim.

2. PaperNexus as Literature-Review Infrastructure for Ideation

Typed review state. PaperNexus starts from indexed papers and semantic snapshots, then commits higher-level review structures into a graph. For ideation, the critical primitives are the domain and mechanism layers plus five typed nodes: `ResearchQuestion`, `Challenge`, `Takeaway`, `IdeaFragment`, and `EvidenceSnippet`. These nodes are not decorative ontology labels. They connect target-domain gaps to mechanism-level abstractions, attach transferable takeaways to their supporting papers, and preserve the spans or snippet references needed for later verification.

Challenge-aware workflow. The repository exposes a dedicated `idea_catalyst` tool whose graph mode accepts a problem statement and target domain and returns either `idea_fragments` or a staged `packet_bundle`. The staged flow encoded in the system is:

1. problem decomposition into target-domain research questions,
2. target-domain analysis of open challenges,
3. cross-domain bridge retrieval over domains, mechanisms, and transferable concepts,
4. source-domain takeaway extraction,

5. recontextualization into target-domain idea fragments, and
6. ranking or requisition.

The public packet contract exposes these stages directly through decomposition, `target_domain_analysis`, `cross_domain_queries`, `source_domain_analyses`, `idea_fragments`, `interdisciplinary_ranking`, and `requisition_report`.

Why this is a review workflow. The core point is not that PaperNexus can “generate ideas.” Many systems can do that. The point is that the proposal surface is constrained by the literature structures already committed to graph state. A candidate idea is therefore not only text. It is a bundle of evidence-carrying objects that can be inspected, compared, downgraded, or rejected before the user treats it as a research direction.

3. Evidence and Failure Boundaries

The main design decision in this workflow is that bridge retrieval alone is insufficient. A cross-domain path can look promising while still being too weak to support an idea fragment. PaperNexus therefore keeps multiple evidence handles attached to intermediate and final outputs: `bridge_path_ids`, `path_trace`, `evidence_chain_refs`, `source_spans`, and an `evidence_tier`. In practical terms, the system distinguishes between “there may be an analogy here” and “there is enough support to surface this as a usable fragment.”

This boundary also structures failure reporting. The current design surfaces at least four failure classes that matter in research use:

- **Alias collision:** mechanism or method normalization can merge incompatible concepts.
- **Weak bridge evidence:** a bridge path exists but has thin source support.
- **Over-proximal domains:** an “interdisciplinary” candidate is only a near-domain restatement.
- **Data starvation:** the graph lacks enough usable support to promote a candidate into an idea fragment.

The most important boundary is the last one. When the graph is data-starved, the system should know that it does not know enough. In PaperNexus, this condition is explicit: `requisition_report.status =`

DATA_STARVATION. This is a stronger contract than simply lowering a confidence score, because downstream users and tools can distinguish “low-ranked idea” from “do not emit an idea yet.”

Equally important, success is defined conservatively. A returned fragment should count as usable only when the packet preserves an auditable bridge path, a non-empty path trace, linked evidence references, at least one source span or evidence snippet, and a non-weak evidence tier. If those conditions are not met, the workflow should fail as a verified ideation workflow even if the prose itself sounds plausible.

4. Evaluation: Retrieval Sanity and Ideation Audit

The evaluation is organized around two different reviewer questions. The first is a standard retrieval question: is the literature-search substrate obviously broken when judged on public retrieval benchmarks? The second is an ideation-audit question: after evidence is retrieved, does the system preserve enough provenance for a reviewer to inspect the bridge claim behind a generated idea? Combining these into one score would be misleading, because ordinary retrieval metrics do not measure source-span preservation, bridge-path evidence, or unsupported cross-domain claims.

4.1. Public Retrieval Benchmarks

Table 1 reports the current retrieval-layer comparison. The BEIR rows use dependency-free BM25 sanity runners on SciFact and NFCorpus (Thakur et al., 2021). The ScholarGym rows use the raw ScholarGym corpus and qrels-backed research queries (Shen et al., 2026); the dense ScholarGym row uses a full-corpus Qwen3-Embedding-0.6B top-100 artifact, and the reranked row applies a BGE cross-encoder over those candidates. The LitSearch rows are the closest match to our workshop use case because they contain natural-language ML/NLP literature-search queries over a static scientific corpus (Ajith et al., 2024). The final LitSearch rows use reciprocal-rank fusion between the PaperNexus lexical ranking and a BGE-small dense ranking artifact (Cormack et al., 2009), followed by a BGE cross-encoder reranking artifact over the top-100 candidates. All rows use standard IR metrics; none evaluates ideation quality.

The main conclusion is not that PaperNexus’ native scorer is a better retriever. It is not. On both LitSearch and ScholarGym, dense retrieval closes a semantic-matching gap that the mostly lexical scorer misses when query language and paper titles have low surface overlap. The useful result is narrower: the benchmark harness can run standard scientific retrieval tasks, including a 570K-paper ScholarGym corpus on the available server, and it can evaluate lexical, dense-

artifact, hybrid, and reranked retrieval under one metric contract. LitSearch and ScholarGym also expose different fusion behavior. On LitSearch, hybrid fusion is the strongest candidate-generation setting for the downstream workflow: compared with lexical scan-50k, it raises Recall@100 from 0.7165 to 0.8126, and reranking raises MRR@10 from 0.3903 to 0.4211. On ScholarGym, however, the Qwen dense row is strongest overall: nDCG@10 rises from the lexical 0.1468 to 0.2225 and Recall@100 from 0.3559 to 0.5357, while RRF and BGE reranking do not improve over dense-only retrieval. This cautions against presenting fusion/reranking as uniformly beneficial; their value depends on the candidate distribution and benchmark query style.

To make the comparison surface less insular, Table 2 lists additional published reference points from the benchmark papers. These rows are not inserted into Table 1 because their protocols differ: LitSearch reports stratified recall at smaller cutoffs, while ScholarGym evaluates an iterative LLM workflow with query planning and relevance assessment rather than a single fixed ranking. They nevertheless locate our results among standard sparse, dense, LLM-reranking, one-hop expansion, and agentic-search baselines.

4.2. Evidence-Contract and Ideation Audit

Table 3 reports the current six-case evidence-contract audit. The settings are: a corrected 7B free-form baseline, a corrected 7B retrieve-then-prompt baseline refreshed with source-span evidence, and PaperNexus catalyst v3. The source artifact is the v6 blinded packet. The labels shown here are from an author-side llama3:latest AI pilot over masked settings. They are useful as a pipeline check, but not as independent human evaluation.

The important update is that source-span packet availability no longer distinguishes PaperNexus from a fair retrieve baseline. Earlier artifacts made retrieve-then-prompt look worse because its prompt-local evidence IDs could not be resolved to paper, span, path, and confidence metadata. After refreshing the retrieve path, retrieve-then-prompt also satisfies the source-span packet contract on all 18 returned items. Therefore the paper cannot claim that PaperNexus uniquely preserves evidence contracts. The remaining question is qualitative and stricter: which setting produces fewer unsupported bridge claims, better target-challenge fit, and more genuinely usable ideas when judged by a calibrated reviewer?

4.3. Controlled Fail-Closed Audit

We retain the controlled fixture because it tests a separate system property. The fixture uses target domain *Education*, source domain *Psychology*, target challenge “reduce confirmation bias during tutoring feedback,” and shared abstract mechanism “metacontrol policy.” It manipulates one sup-

Table 1. Retrieval-layer sanity results. These rows are comparable as retrieval benchmarks, but they do not measure evidence-contract preservation or idea quality.

Benchmark	Method	Docs	Queries	nDCG@10	Recall@100	MRR@10
BEIR SciFact	BM25 over title+text, dependency-free runner	5,183	300	0.6622	0.8826	0.6302
BEIR NFCorpus	BM25 over title+text, dependency-free runner	3,633	323	0.3073	0.2359	0.5138
ScholarGym	Pruned BM25 over title+abstract	570,206	2,458	0.1526	0.3671	0.1546
ScholarGym	PaperNexus fixed-corpus scorer, scan 10k	570,206	2,458	0.1470	0.3507	0.1529
ScholarGym	PaperNexus fixed-corpus scorer, scan 50k	570,206	2,458	0.1468	0.3559	0.1535
ScholarGym	Qwen3-Embedding-0.6B dense artifact	570,206	2,458	0.2225	0.5357	0.2289
ScholarGym	PaperNexus hybrid RRF, lexical+dense	570,206	2,458	0.2044	0.5265	0.2131
ScholarGym	BGE reranker over Qwen top-100	570,206	2,458	0.2092	0.5357	0.2093
LitSearch	PaperNexus lexical scorer, scan 50k	64,183	597	0.3739	0.7165	0.3333
LitSearch	BGE-small dense artifact, same evaluator	64,183	597	0.4343	0.7864	0.3943
LitSearch	PaperNexus hybrid RRF, lexical+dense	64,183	597	0.4356	0.8126	0.3903
LitSearch	PaperNexus hybrid RRF + BGE reranker	64,183	597	0.4621	0.8126	0.4211

port signal: the presence of a source `EvidenceSnippet`. With the snippet present, the implementation returns bridge paths, source-domain analyses, evidence-chain references, source spans, and at least one idea fragment. When the snippet is removed, bridge retrieval still finds candidate paths, but no usable fragment is emitted and the requisition report marks `DATA_STARVATION`.

This fail-closed behavior is still the clearest positive system property: evidence should precede ideation, not merely decorate generated prose after the fact.

5. Discussion and Reproducibility

PaperNexus should be read as infrastructure, not autonomous science. The graph-mode workflow contributes a disciplined intermediate layer between literature review and research ideation. It does not replace researcher judgment, and it does not yet establish end-task gains such as better papers or faster discoveries.

The updated benchmark results sharpen the claim. At the retrieval layer, PaperNexus is comparable enough to run public scientific-search tasks, but the native fixed-corpus scorer alone is not competitive with dense retrieval on the closest public benchmarks. This is an engineering gap, not a rhetorical detail: lexical ranking depends on scan limits and can miss relevant papers when the query asks for “challenges,” “extensions,” or “applications” while candidate titles use semantically related but lexically different language. The practical repair path is to keep PaperNexus’ auditable evaluator and graph-facing workflow, but feed it semantic retrieval signals before synthesis. Fusion and reranking should remain measured components rather than assumed improvements: they help LitSearch, but the full ScholarGym run shows that a strong dense first stage can outperform the hybrid and reranked variants.

At the ideation layer, the benchmark gap is different. The v6 refresh shows that a fair retrieve-then-prompt baseline can

also carry source-span packets. That invalidates the simple claim that only PaperNexus preserves evidence metadata. The final comparison must therefore be judged by stricter semantic labels: whether the proposed bridge is actually supported, whether the idea addresses the target challenge, whether OCR or noisy snippets drive the claim, and whether the fragment is usable for a reviewer-facing result. The current AI-pilot labels prove the labeling pipeline works, but they are too permissive to substitute for independent human judgment.

This gives a more honest workshop positioning. Compared with literature-search systems, PaperNexus’ added value is not better first-stage retrieval by default; the strongest retrieval row requires adding dense evidence, and the best fusion/reranking choice is benchmark-dependent. Compared with free-form brainstorming, its added value is not demonstrated novelty. The credible claim is that research ideation can be routed through auditable review objects and explicit refusal boundaries. The current artifacts support that workflow claim and expose where the system fails: native retrieval needs semantic integration, reranking needs benchmark-specific validation, and idea-quality claims still need stricter adjudication.

Reproducibility is close to the implementation surface. The repository contains: (i) the staged skill specification for the catalyst workflow, (ii) the graph design notes for cross-domain intelligence, (iii) benchmark loaders and fixed-corpus retrieval runners, (iv) v6 masked adjudication packets, and (v) scripts for AI-pilot labeling and aggregation. Appendix A gives a minimal invocation and correctness protocol. The supplementary package should be treated as an artifact-backed workflow audit with public retrieval sanity checks, not as a completed human-evaluation benchmark.

References

Ajith, A., Xia, M., Chevalier, A., Goyal, T., Chen, D., and Gao, T. Litsearch: A retrieval benchmark for scientific

Table 2. Expanded published reference baselines from the underlying benchmark papers (Ajith et al., 2024; Shen et al., 2026). These numbers are protocol references, not strict apples-to-apples rows with Table 1. LitSearch reports average broad R@20 / average specific R@5. ScholarGym reports Test-Fast retrieval recall / selected-paper F1 under sparse abstract-only retrieval at $T = 5$, except direct-query rows at $T = 1$.

Benchmark	Published method	Protocol	Reported score(s)	Role in comparison
LitSearch	BM25	title+abstract retrieval	39.9 / 50.0	Sparse lexical reference
LitSearch	GTR-T5-large	title+abstract dense retrieval	43.8 / 39.6	Dense encoder reference
LitSearch	Instructor-XL	title+abstract dense retrieval	56.5 / 52.3	Instruction-tuned dense reference
LitSearch	E5-large-v2	title+abstract dense retrieval	55.4 / 56.2	Strong dense reference
LitSearch	GritLM-7B	title+abstract dense retrieval	70.8 / 74.8	Strongest reported retriever row
LitSearch	GPT-4o rerank w/ BM25	top-100 LLM reranking	59.9 / 68.0	LLM reranking from sparse candidates
LitSearch	GPT-4o one-hop w/ BM25	one-hop citation expansion	64.8 / 68.6	Query expansion from sparse candidates
LitSearch	GPT-4o rerank w/ GritLM	top-100 LLM reranking	75.3 / 79.2	Strongest reported reranking row
LitSearch	GPT-4o one-hop w/ GritLM	one-hop citation expansion	73.2 / 77.0	Expansion from strong dense candidates
ScholarGym	Direct Query, Qwen3-8B	Test-Fast, sparse, $T = 1$	0.210 / 0.069	Small open direct-query baseline
ScholarGym	Direct Query, Qwen3-30B	Test-Fast, sparse, $T = 1$	0.350 / 0.098	Larger direct-query baseline
ScholarGym	Qwen3-8B workflow	Test-Fast, sparse, $T = 5$	0.550 / 0.231	Small open agentic workflow
ScholarGym	Qwen3-8B [†] workflow	Test-Fast, sparse, $T = 5$	0.580 / 0.293	Small open thinking workflow
ScholarGym	Qwen3-30B workflow	Test-Fast, sparse, $T = 5$	0.720 / 0.285	Larger open workflow
ScholarGym	Qwen3-30B [†] workflow	Test-Fast, sparse, $T = 5$	0.590 / 0.362	Best open-source F1 row
ScholarGym	GLM-4.7 workflow	Test-Fast, sparse, $T = 5$	0.814 / 0.194	High-recall open/proprietary reference
ScholarGym	DeepSeek-V3.2 workflow	Test-Fast, sparse, $T = 5$	0.862 / 0.233	Proprietary workflow reference
ScholarGym	DeepSeek-V3.2 [†] workflow	Test-Fast, sparse, $T = 5$	0.872 / 0.423	Extended-thinking workflow reference
ScholarGym	GPT-5.2 workflow	Test-Fast, sparse, $T = 5$	0.883 / 0.447	Highest reported selected-paper F1
ScholarGym	Gemini3-Pro workflow	Test-Fast, sparse, $T = 5$	0.958 / 0.329	Highest reported retrieval recall

Table 3. Six-case ideation audit status. The AI-pilot label columns are diagnostic only; the final ideation table still needs independent or separately calibrated labels.

Setting	Items	Packet pass	Grounded yes/partial	Unsupported yes	Usable
Corrected 7B baseline	18	0/18	0/18	18/18	0/18
Retrieve-then-prompt, source-span refreshed	18	18/18	17/18	1/18	17/18
PaperNexus catalyst v3	10	10/10	8/10	2/10	8/10

Table 4. Controlled packet-bundle audit. Removing evidence preserves possible bridge structure but suppresses user-facing fragments.

Condition	Paths	Trace	Spans	Fragments	Outcome
Snippet kept	>0	>0	>0	≥1	fragment
Snippet removed	>0	>0	0	0	requisition

literature search. <https://arxiv.org/abs/2407.18940>, 2024.

Asai, A., He, J., Shao, R., Shi, W., Singh, A., Chang, J. C., Lo, K., Soldaini, L., Feldman, S., D’arcy, M., Wadden, D., Latzke, M., Tian, M., Ji, P., Liu, S., Tong, H., Wu, B., Xiong, Y., Zettlemoyer, L., Neubig, G., Weld, D., Downey, D., Yih, W.-t., Koh, P. W., and Hajishirzi, H. Openscholar: Synthesizing scientific literature with retrieval-augmented language models. <https://arxiv.org/abs/2411.14199>, 2024.

Cormack, G. V., Clarke, C. L. A., and Buettcher, S. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009.

Kargupta, M., Park, J., Lin, X., McAuley, J., Burdick, J., and Marlin, B. M. Sparking scientific creativity via llm-driven interdisciplinary inspiration. <https://arxiv.org/abs/2603.12226>, 2026.

Lála, J., O’Donoghue, O., Shtedritski, A., Cox, S., Rodrigues, S. G., and White, A. D. Paperqa: Retrieval-augmented generative agent for scientific research. <https://arxiv.org/abs/2312.07559>, 2023.

Shen, H., Yang, H., Gu, Z., and Han, W. ScholarGym: Benchmarking large language model capabilities in the information-gathering stage of deep research. <https://arxiv.org/abs/2601.21654>, 2026.

Thakur, N., Reimers, N., Rücklé, A., Srivastava, A., and Gurevych, I. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. <https://arxiv.org/abs/2104.08663>, 2021.

Wu, Y., Zhang, D., Li, X., Xu, J., Duan, Y., Liu, Y., Pan, J., Zhu, Q., Zhou, X., Wei, J., Li, S., Chen, J., He, C., and Tan, C. Intern-atlas: A methodological evolution graph as research infrastructure for ai scientists. <https://arxiv.org/abs/2604.28158>, 2026.

Xiong, L., Luo, K., Xia, Z., Zhang, W., Yao, J.-G., Liu, Z., Shao, J., Chen, J., Qian, H., Yang, X., Yu, Q., Li, H., Yue, C., Du, X., Wang, Y., Liu, Y., Xu, H., and Dou, Z. Autoresearchbench: Benchmarking ai agents on complex scientific literature discovery. <https://arxiv.org/abs/2604.25256>, 2026.

A. Minimal Reproduction and Correctness Checks

Minimal command. The repository already exposes a CLI entry point for the same target-domain challenge used in the controlled fixture:

```
papernexus catalyst --target-domain Education \
  --challenge "reduce confirmation bias during tutoring feedback"
```

Implementation artifacts. The current draft is grounded in three repository artifacts: the staged workflow description in SKILL/PaperNexusIdeaCatalyst/SKILL.md, the graph-layer notes in docs/graph/cross-domain-intelligence.md, and the packet-bundle audit in test/idea-catalyst-packets.test.js. The public MCP description additionally documents that idea-catalyst returns either idea fragments or a staged packet bundle.

Baseline prompt skeleton. The comparison against free-form ideation uses a fixed prompt template rather than ad hoc prompting. A minimal version is:

```
You are helping with early-stage research ideation.
Target domain: <target domain>
Research challenge: <challenge statement>
```

```
Propose 3 cross-domain research ideas.
For each idea, give:
- source domain
- transferable mechanism
- short rationale
- what evidence would still need checking
```

Do not invent citations or claim evidence you cannot support.

This is a template for reproducibility, not a claimed transcript from a completed benchmark run.

Packet-bundle excerpt. The graph-grounded path should save the returned packet bundle. A minimally useful audit excerpt, derived from the tested public contract, has the following shape:

```
{
  "decomposition": {"questions": [...]},
  "target_domain_analysis": [...],
  "source_domain_analyses": [{
    "bridge_path_ids": [...],
    "path_trace": [...],
    "evidence_chain_refs": [...],
    "source_spans": [...]
  }],
  "idea_fragments": [{
    "title": "...",
    "bridge_path_ids": [...],
    "evidence_tier": "moderate"
  }],
  "requisition_report": null
}
```

If the graph is too weakly supported, the packet should instead contain an empty idea_fragments list and a non-null requisition_report with DATA_STARVATION.

Minimal correctness protocol. For each returned idea fragment, a verifier should check:

1. at least one attached `bridge_path_id`,
2. at least one non-empty `path_trace`,
3. at least one `evidence_chain_ref`,
4. at least one usable `source_span` or linked evidence snippet, and
5. a non-weak evidence tier.

If these conditions fail, the workflow should not emit a usable fragment. Instead, it should return a requisition with missing evidence types, as in the controlled starvation audit.

Task contract. For this paper, the intended input-output contract can be summarized as:

target-domain challenge $\rightarrow$ evidence-backed idea fragments *or* requisition

This compact contract is useful when evaluating new cases: if the system returns unsupported prose, the workflow has failed; if it returns a requisition when evidence is missing, the refusal boundary has behaved as intended.

Scope note. This paper intentionally evaluates only the repository’s graph-grounded mode. PaperNexus also contains a live-discovery variant aligned with the Idea-Catalyst paper, but that online path is supplementary to the main claim here. The contribution of the present draft is the evidence-preserving graph workflow and its explicit refusal boundary.

Transcript expectations. For each future case, the supplementary package should preserve four small artifacts: the input challenge statement, the baseline prompt and response, the catalyst packet bundle, and the reviewer annotation sheet. That is enough to rerun the comparison and audit correctness without turning the workshop submission into a large benchmark appendix. In the current repository layout, these materials are scaffolded by the draft-local files `evaluation-cases.md`, `baseline-prompt-template.txt`, `packet-bundle-excerpt.json`, `annotation-template.csv`, `transcript-bundle-checklist.md`, and `evaluation-runbook.md`, plus the concrete per-case directories under `evaluation-artifacts/`.

Archived live pilot. The draft-local supplementary package now also includes one non-counted pilot under `evaluation-artifacts/case-00-live-gcd-domain-shift/`. It stores a real graph-mode packet from the only currently available indexed local corpus, a manual annotation that marks the returned `DATA_STARVATION` as correct, and one archived free-form baseline transcript for the same challenge. This remains a qualitative contrast rather than a counted benchmark result, but it now demonstrates that the supplementary package can preserve both sides of a baseline-versus-graph comparison.

Counting rules and case maturity. To keep the workshop draft honest, the supplementary files include a case manifest and a v6 masked adjudication packet. Cases 01–06 now have archived corrected 7B baseline outputs, refreshed retrieve-then-prompt outputs with source-span packets, and PaperNexus catalyst v3 outputs. They are mechanically aggregate-ready, but the current aggregate labels are author-side AI-pilot labels. The final paper table should remain provisional until independent human labels or a separately calibrated adjudicator have been applied to `masked-adjudication-items-v6.jsonl`.

Controlled fixture audit. The positive-versus-starvation packet-bundle audit used in the main text is now also restated in `controlled-fixture-audit.md`. That note maps the repository test fixture to the two claims actually used here: snippet present implies auditable spans and at least one usable fragment, while snippet removed preserves possible bridge structure but suppresses fragments and returns `DATA_STARVATION`.

Failure-mode examples. Table 5 summarizes the current qualitative failure taxonomy together with the evidence handle that should let a reviewer detect each failure.

Table 5. Current failure modes and the evidence handle expected to surface them.

Failure mode	Typical cause	Expected evidence handle
Alias collision	normalization merges distinct mechanisms or methods	conflicting path trace or mismatched supporting nodes
Weak bridge evidence	analogy exists but source support is thin	sparse evidence refs, weak evidence tier, or missing source spans
Over-proximal domains	candidate is near-domain reformulation rather than real transfer	low domain-distance signal and low interdisciplinary ranking diversity
Data starvation	graph lacks enough support for a usable fragment	empty idea fragments plus explicit DATA_STARVATION requisition

Table 6. Current claim boundary for the workshop draft.

Item	Status in this draft	Current evidence basis
Evidence-preserving ideation workflow	claimed	repository workflow contract, packet schema, controlled fixture audit
Fail-closed DATA_STARVATION boundary	claimed	controlled snippet-removal fixture plus one archived live pilot starvation packet
Improved grounded idea rate over baseline	not yet claimed	v6 AI-pilot labels exist, but independent labels are still missing
Reduced unsupported bridge claims in the wild	not yet claimed	v6 AI-pilot labels exist, but the strict human-calibration gap remains
Researcher time savings or paper-quality gains	not yet claimed	no user study or time audit has been run
Retrieval superiority over scholarly assistants	not claimed	outside the scope of the current evidence

Claim boundary matrix. Table 6 summarizes what this draft does and does not currently claim, together with the evidence basis for each item. The point is to keep the workshop version narrow and auditable.

B. Minimum Case Set for the Next Evaluation

Why this appendix exists. The main paper currently reports public retrieval sanity results and an artifact-backed six-case ideation audit, not a completed human user study. This appendix fixes the minimum reproducible evaluation package and clarifies which parts are final artifacts versus provisional labels.

Recommended six-case inventory. Each case is phrased as a target-domain challenge that plausibly benefits from cross-domain transfer while remaining auditable by a human reviewer.

1. **Model editing robustness:** how can model editing methods avoid unstable local fixes that degrade behavior elsewhere?
2. **Long-context evaluation reliability:** how can evaluation protocols better detect brittle long-context reasoning rather than superficial retrieval success?
3. **Preference optimization under sparse feedback:** how can alignment methods remain stable when preference labels are noisy or limited?
4. **Retrieval-augmented reasoning with weak supervision:** how can systems improve evidence use when relevance labels are scarce or incomplete?
5. **Tool-use planning under uncertainty:** how can agentic systems decide when to defer, ask, or stop rather than over-commit to a bad plan?
6. **Continual adaptation under shifting user intent:** how can human-AI systems adapt to evolving goals without reinforcing stale assumptions?

Comparison settings. The smallest honest comparison uses three settings:

1. **Baseline free-form ideation:** prompt an LLM directly with the target-domain challenge and ask for cross-domain ideas.

2. **Retrieve then prompt:** use the same challenge together with top- k raw title/abstract/snippet evidence returned by the existing query surface on the same fixed corpus, but without graph bridge traces or packet structure.
3. **PaperNexus graph-grounded catalyst:** run the graph-mode `idea_catalyst` workflow and inspect the returned packet bundle or idea fragments.

Annotation rubric. For each returned idea fragment or baseline idea, a human reviewer should record:

1. whether the idea is **grounded** in at least one identifiable source span or evidence reference,
2. whether any **bridge claim** is unsupported by the attached evidence,
3. whether the returned idea covers a **real target challenge** rather than a nearby reformulation,
4. whether the packet preserves enough structure for manual verification, and
5. whether the system correctly emits `DATA_STARVATION` when support is too weak.

Adjudication and aggregation. For a counted workshop table, the first three metrics should be annotated at the returned-idea level, while starvation should be judged at the case-output level. Each baseline idea or PaperNexus idea fragment should therefore receive its own row in the annotation sheet; explicit requisition-only outputs should receive one row with a blank idea id. Aggregation should use only cases marked `aggregate_eligible=yes` in the case manifest. Grounded idea rate is the fraction of returned-idea rows marked grounded, unsupported claim rate is the fraction marked unsupported, evidence coverage is the fraction marked sufficient, and starvation precision is the fraction of explicit starvation outputs judged correct.

Counted-case readiness. One workshop case should become `aggregate_eligible` only after seven conditions are met: a queryable corpus has been assigned, the challenge statement is frozen, the baseline output is archived, the graph-grounded packet is archived, the annotation sheet has been expanded to one row per returned idea or explicit starvation output, the calibration pass has been completed, and any required adjudication has been closed. Until then, the case belongs to the reproducibility scaffold but not to the counted result table.

Reviewer calibration and bias control. Because the counted workshop comparison is small, annotation drift can dominate the apparent result. The draft-local protocol therefore recommends a short calibration pass on a non-counted case, fixed annotation order, and condition masking when feasible. These steps do not make the evaluation fully blind, but they reduce the risk that baseline-versus-PaperNexus expectations leak directly into row labels.

Minimum result table. The first real evaluation table should contain only four semantic/audit metrics in addition to packet availability. The current v6 AI-pilot version is:

Setting	Grounded	Unsupported	Coverage	Starvation
Baseline LLM	0/18	18/18	0/18	n/a
Retrieve+prompt	17/18	1/18	18/18	n/a
PaperNexus	8/10	2/10	10/10	n/a

Here “Grounded” is grounded yes or partial, “Unsupported” is unsupported bridge-claim rate, “Coverage” is source-span or evidence-ref coverage, and “Starvation” is starvation precision. These numbers should be labeled as AI-pilot diagnostics, not final independent labels.